**Math 135** **Homework 5**
**UCLA, Fall 2025** **due 11/16/2025 at 11:59pm**

**Directions.** Complete all the problems. Your solutions to the problems should be submitted to Gradescope before the indicated due date above. Please follow rules regarding Gradescope submission as described in the syllabus. If you use any outside source in your solutions, you must cite the source.

**Problem 1.** Suppose that $f : \mathbb{R} \to \mathbb{R}$ is continuously differentiable. Prove that for any interval $I \subseteq \mathbb{R}$ of bounded length, $f$ restricted to domain $I$ is Lipschitz.

*Hint.* You may use the Extreme Value Theorem from calculus.

**Solution.** This just requires a small modification of the argument we gave in lecture.

$I$ is an interval of bounded length, which is possible missing one or both of its endpoints. Let $\overline{I}$ be the closed and bounded interval obtained by adding to $I$ any of its missing endpoints. Then, $f'$ is a continuous function on the closed and bounded interval $\overline{I}$, hence by the Extreme Value Theorem $f'$ is bounded on $\overline{I}$. Then, the argument proceeds as before, using the Mean Value Theorem to show that any bound for $|f'|$ on $\overline{I}$ is a Lipschitz constant for $f$ restricted to $I$.

**Problem 2.** Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^2$. In this problem, you will show that $f$ is not Lipschitz (on domain $\mathbb{R}$).

(a) Prove that for any $L > 0$, there exists $x, y \in \mathbb{R}$ such that $|x^2 - y^2| = (L+1)|x - y|$.

*Hint.* Take $x = 1$ and $y = L$. Also, remember how to factor the difference of two squares: $x^2 - y^2 = (x - y)(x + 1)$.

(b) Explain how (a) implies that $f$ is not Lipschitz on its domain, $\mathbb{R}$.

(c) If we restrict $f$ to any finite length interval $I$, is $f$ Lipschitz on $I$? Explain your answer.

**Solution.** (a) Just follow the hint. Fix $L > 0$. Let $x = 1$ and $y = L$. Then,

$$|x^2 - y^2| = |x + y||x - y| = |L + 1||x - y| = (L + 1)|x - y|.$$

(b) Suppose towards a contradiction that $f$ is Lipschitz. Let $L$ be a Lipschitz constant for $f$. Note that we can pick $L > 1$ since any number bigger than a Lipschitz constant is also a Lipschitz constant. Then, taking $x = 1$ and $y = L$ we have

$$(L+1)|x-y| = |x^2 - y^2| = |f(x) - f(y)| \leq L|x-y|.$$

Since $L > 1$, $|x - y| \neq 0$, so we can divide both sides of the inequality above by $|x - y|$. This implies that $L + 1 \leq L$, a contradiction.

(c) Yes. This follows immediately from the previous problem.

**Problem 3.** Let $a < b$ and let $D \subseteq \mathbb{R}$. Show that if $g(t)$ is bounded on $[a, b]$ and $h(y)$ is Lipschitz on $D$, then $f(t, y) = g(t)h(y)$ is Lipschitz in $y$ on $[a, b] \times D$.

**Solution.** Let $L_h$ be a Lipschtz constant for $h$ on $D$. Let $M > 0$ be a constant such that $|g(t)| \leq M$ for all $a \leq t \leq b$. Then for any $y_1, y_2 \in D$ and any $a \leq t \leq b$,

$$|f(t,y_1) - f(t,y_1)| = |g(t)h(y_1) - g(t)h(y_2)| = |g(t)||h(y_1) - h(y_2)| \leq ML_h|y_1 - y_2|.$$

Thus, $ML_h$ is a Lipschitz constant for $f(t, y)$ in $y$ on $\Omega$.

**Problem 4.** For the initial value problem

$$y' = y^3 + 2, \qquad y(0) = 1,$$

show that there is some interval $I$ with $0 \in I$ such that the IVP has a unique solution defined on $I$.

**Solution.** We write the IVP in the form

$$y' = f(t, y), \qquad y(0) = 1,$$

where $f : [-1, 1] \times [0, 2] \to \mathbb{R}$ is defined by $f(t, y) = y^3 + 2$. Note that our initial condition pair $(0, y(0)) = (0, 1)$ is in the domain of $f$. Now, by problems above, $y^3 + 2$ restricted to $[0, 2]$ is Lipschitz since it is continuously differentiable. Thus, $f$ is Lipschitz in $y$ on $[-1, 1] \times [0, 2]$. Thus, we can apply the weaker statement of Picard-Lindelof to guarantee a unique solution on a time interval of the form $[-\delta, \delta]$ for some $\delta > 0$.

**Problem 5.** Show that the initial value problem

$$y' = y + \cos y, \qquad y(0) = 1$$

has a unique solution on any interval of the form $[-M, M]$, where $M > 0$.

**Solution.** Fix $M > 0$. Write our IVP in the form

$$y' = f(t, y), \qquad y(0) = 1,$$

where $f : [-M, M] \times \mathbb{R} \to \mathbb{R}$ is defined by $f(t, y) = y + \cos y$. The function $y + \cos y$ has derivative $1 - \sin(y)$, which is bounded on all of $R$ since

$$|1 - \sin(y)| \le 2 \qquad \text{for all } y \in \mathbb{R}$$

Thus, $f(t, y)$ is Lipschitz in $y$ on $[-M, M] \times \mathbb{R}$. The desired result now follows from Picard-Lindelof.

**Problem 6.** (a) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function $f(t, y) = t^2 y$. Show that for any $a > 0$ the function $f$ is Lipschitz in $y$ on $\Omega = [-a, a] \times \mathbb{R}$.

(b) Solve the IVP $y' = t^2 y$, $y(0) = 1$. Is the solution you found the only solution? Justify your answer.

(c) Let $f(t, y) = ty^2$. Show that for any $c < d$ the function $f$ is Lipschitz in $y$ on $\Omega = [-1, 1] \times [c, d]$.

(d) Solve $y' = ty^2$, $y(0) = 2$. Can you find a solution to this IVP which is defined on all of $[-1, 1]$? How does the answer to that question relate to whether $f(t, y) = ty^2$ is Lipschitz on $[-1, 1] \times \mathbb{R}$?

**Solution.** (a) Fix $a > 0$. Let $g(t) = t^2$ and $h(y) = y$. Then $f(t, y) = g(t)h(y)$, with $g(t)$ obviously bounded on $[-a, a]$, and $h(y)$ is Lipschitz on $\mathbb{R}$ (very easy to show directly; we did it lecture). It follows from a problem above that $f(t, y)$ is Lipschitz in $y$ on $[-a, a] \times \mathbb{R}$.

(b) The solution to the IVP is $y(t) = e^{t^3/3}$, $-\infty < t < \infty$. (Details skipped here, but can be solved by separating the variables)

This solution is unique. Suppose by contradiction that $\tilde{y} : \mathbb{R} \to \mathbb{R}$ is different solution. Then, there is some $a > 0$ such that $y(t) = e^{t^2/2}$ and $\tilde{y}(t)$ differ on $[-a, a]$. By the result of part (a), $f(t, y)$ is Lipschitz on $[-a, a] \times \mathbb{R}$. Thus, by Picard-Lindelöf, the IVP has a unique solution defined on $[-a, a]$. Since both $y(t)$ and $\tilde{y}(t)$ (restricted to $[-a, a]$) are solutions to the IVP, they must be the same on $[-a, a]$. This contradicts our choice of $a$, i.e., that $y(t) = e^{t^2/2}$ and $\tilde{y}(t)$ differ on $[-a, a]$.

(c) Fix $c < d$. Let $g(t) = t$ and $h(y) = y^2$. Then $f(t, y) = g(t)h(y)$, with $g(t)$ bounded on $[-1, 1]$, and $h(y)$ is Lipschitz on $[c, d]$ (since its derivative is bounded on $[c, d]$). It follows from previous problem that $f(t, y)$ is Lipschitz in $y$ on $[-1, 1] \times [c, d]$.

(d) The solution is $y(t) = \frac{2}{1-t^2}$, $t \neq -1, 1$. This solution is not defined at $-1$ or $1$. $f(t, y) = ty^2$ is not Lipschitz in $y$ on $[-1, 1] \times \mathbb{R}$, so Picard-Lindelöf does not guarantee the existence of a unique solution on all of $[-1, 1]$. In part (c), we showed that $f(t, y)$ is Lipschitz in $y$ on $[-1, 1] \times [2 - c, 2 + c]$, for any choice of $c > 0$. Thus, Picard-Lindelöf only gurantees the existence of a solution defined on some $[-\delta, \delta]$, with $0 < \delta < 1$, which from our solution above seems to be the strongest possible conclusion.

**Problem 7.** Consider the initial value problem

$$y' = \frac{2y}{t-1}, \qquad y(0) = 1.$$

(a) Use Picard-Lidelöf (or its corollary) to show that there is $\delta > 0$ such that the IVP has a unique solution $y : [-\delta, \delta] \to \mathbb{R}$.

(b) Verify that for each $c \in \mathbb{R}$,

$$y(t) = \begin{cases} (t-1)^2 & \text{if } t < 1 \\ c(t-1)^2 & \text{if } t \geq 1 \end{cases}$$

is a solution to the IVP.

(c) Use part (b) to discuss the possible value of the $\delta > 0$ from part (a).

**Solution.** (a) Consider $\Omega = [-1/2, 1/2] \times \mathbb{R}$. Let $g(t) = 2/(t-1)$ and let $h(y) = y$. Since $g(t)$ is continuous on $[-1/2, 1/2]$, it is also bounded on $[-1/2, 1/2]$ by the extreme value theorem. $h(y)$ is Lipschitz on $\mathbb{R}$. Thus, $f(t, y) = g(t)h(y)$ is Lipschitz in $y$ on $\Omega$ by a previous problem. Thus, Picard-Lindelöf guarantees the existence of a unique solution $y : [-1/2, 1/2] \to \mathbb{R}$ to the IVP.

(b) Straightforward.

(c) The largest possible value of $\delta$ is $\delta = 1$. This is because part (b) shows us that once $\delta > 1$, we have many solutions on $[-\delta, \delta]$

**Problem 8.** For $y_0 \geq 0$, consider the IVP

$$y' = 5t^3 y^{1/5}, \qquad y(0) = y_0. \tag{$*$}$$

(a) Let $y_0 > 0$. Prove that there is $\delta > 0$ so that $(*)$ has a unique solution defined on $[-\delta, \delta]$.

(b) Now, let $y_0 = 0$. Show that $y(t) = 0$ for all $t$ is a solution to $(*)$. Then, use separation of variables to find another solution to $(*)$.

(c) Is the function $f(t, y) = 5t^3 y^{1/5}$ Lipschitz in $y$ on $[0, 1] \times [0, 1]$? Explain your answer.

**Solution.** (a) Let $g(t) = 5t^3$ and $h(y) = y^{1/5}$, so that $f(t, y) = g(t)h(y)$. Pick $c, d$ with $0 < c < y_0 < d$. By the usual arguments used above, $f$ is Lipschitz in $y$ on $[-1, 1] \times [c, d]$. Then, we can apply Picard-Lindelof (weaker statement) to get the desired result.

(b) The other solution is $y(t) = t^5$.

(c) No, it's not. If it were, we could apply Picard-Lindelof to guarantee a unique solution to the IVP

$$y' = 5t^3 y^{1/5}, \qquad y(0) = 0$$

on some $[-\delta, \delta]$. But we found in part (b) two solutions that differ on any interval of that form.